

Probability and Statistics - XB_0115

Chapter 7 - Zoe Zhou

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Law of Large Numbers (LLN)

Definition: Let $X_1, \dots, X_n$ be iid random variables with finite mean μ . Then for any $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq \epsilon) = 0$$

In plain words...

- As you collect more data, the **sample average** ($\bar{X}_n$) gets closer and closer to the **true population mean** (μ).
- Individual flukes and outliers eventually "cancel out."
- *Example:* If you flip a fair coin 10 times, you might get 70% heads. If you flip it 10,000 times, you will almost certainly be very close to 50%.

Central Limit Theorem (CLT)

Definition: Let $X_1, \dots, X_n$ be iid with mean μ and variance σ^2 , and let $Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$.

Then for any $x \in \mathbb{R}$:

$$\lim_{n \rightarrow \infty} P(Z_n \leq x) = \Phi(x)$$

where Φ is the CDF of the standard normal distribution $N(0, 1)$.

Key Properties:

- Note that $E(Z_n) = 0$ and $\text{Var}(Z_n) = 1$.
- **The Magic:** It doesn't matter what the original distribution of X_i looks like (it could be skewed, uniform, or discrete); the **sum** (or average) will always look like a Bell Curve if n is large enough ($n \geq 30$ is the usual rule of thumb).

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Problem Statement

1. Let X_i be i.i.d $Uniform(0, 1)$. We define the sample mean as

$$M_n = \frac{X_1 + X_2 + \dots + X_n}{n}.$$

- (a) Find $E[M_n]$ and $\text{Var}(M_n)$ as a function of n .
(b) Using the Chebyshev's inequality, find an upper bound on

$$P\left(\left|M_n - \frac{1}{2}\right| \geq \frac{1}{100}\right).$$

- (c) Using your bound, show that

$$\lim_{n \rightarrow \infty} P\left(\left|M_n - \frac{1}{2}\right| \geq \frac{1}{100}\right) = 0.$$

Part (a): Expectation and Variance of M_n

Uniform Distribution, Expectation and Variance Properties

For $X_i \sim \text{Uniform}(a, b)$, the individual mean and variance are:

$$E[X_i] = \frac{a+b}{2}, \quad \text{Var}(X_i) = \frac{(b-a)^2}{12}$$

For independent random variables $X_1, \dots, X_n$:

$$E \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i], \quad \text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i)$$

Part (a): Solution

Expectation of M_n :

$$E[M_n] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{n \cdot (1/2)}{n} = \frac{1}{2}$$

Variance of M_n : Since X_i are independent:

$$\text{Var}(M_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{n \cdot (1/12)}{n^2} = \frac{1}{12n}$$

Part (b): Chebyshev's Inequality

Chebyshev's Inequality

For any random variable X with mean μ and variance σ^2 :

$$P(|X - \mu| \geq \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2}$$

In plain words: Chebyshev's Inequality is used to find the **upper bound** of the probability. No matter what the distribution looks like, the probability of getting an "extreme" value (far from the average) cannot exceed this bound.

To find the bound for $P(|M_n - \frac{1}{2}| \geq \frac{1}{100})$:

- We have $\mu = E[M_n] = \frac{1}{2}$ and $\epsilon = \frac{1}{100}$.
- Substituting $\text{Var}(M_n) = \frac{1}{12n}$:

$$P\left(|M_n - \frac{1}{2}| \geq \frac{1}{100}\right) \leq \frac{\text{Var}(M_n)}{(\frac{1}{100})^2} = \frac{1/12n}{1/10000} = \frac{10000}{12n}$$

Part (c): Squeeze Theorem and Convergence

The Squeeze Theorem Logic

If $0 \leq P(A) \leq f(n)$ and $\lim_{n \rightarrow \infty} f(n) = 0$, then the probability $P(A)$ must also converge to 0 because probability is non-negative.

Using the upper bound from Part (b):

$$0 \leq P\left(\left|M_n - \frac{1}{2}\right| \geq \frac{1}{100}\right) \leq \frac{10000}{12n}$$

Taking the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \frac{10000}{12n} = 0$$

Meaning

As n increases, the probability that the sample mean deviates from the true mean by more than 0.01 vanishes. This is a direct proof of the Weak Law of Large Numbers (WLLN).

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Problem Statement

2. The number of accidents in a certain city is modeled by a Poisson random variable with average rate of 10 accidents per day. Suppose that the number of accidents in different days are independent. Use the central limit theorem to find the probability that there will be more than 3800 accidents in a certain year. Assume that there are 365 days in a year.

Part (a): Modeling the Accident Rate

Why Poisson Distribution?

The Poisson distribution is ideal for modeling the number of times an event occurs in a **fixed interval of time** (like a day), provided:

- Events occur independently (one accident doesn't cause another).
- The average rate (λ) is constant over time.

From the Problem Statement:

- The "average rate" is given as 10 accidents per day.
- In Poisson notation, this means $\lambda = 10$.
- Therefore, for any single day i : $X_i \sim \text{Poisson}(10)$.

Key Property

A unique feature of the Poisson distribution is that its **Mean** and **Variance** are identical:

$$E[X_i] = \text{Var}(X_i) = \lambda = 10$$

Part (b): Applying Central Limit Theorem

Central Limit Theorem (CLT)

When n is large, the sum Y of i.i.d. variables follows a normal distribution:

$$\frac{Y - E[Y]}{\sqrt{\text{Var}(Y)}} \approx N(0, 1)$$

Normalization Step: We want to find $P(Y \geq 3800)$. We transform Y into a standard normal variable Z :

$$P(Y \geq 3800) = P\left(\frac{Y - 3650}{\sqrt{3650}} \geq \frac{3800 - 3650}{\sqrt{3650}}\right)$$

Part (c): Final Probability Calculation

Complement Rule for CDF

Since $\Phi(x)$ gives $P(Z \leq x)$, the upper tail probability is:

$$P(Z \geq x) = 1 - \Phi(x)$$

Calculation:

$$\begin{aligned} P\left(Z \geq \frac{150}{\sqrt{3650}}\right) &\approx 1 - \Phi(2.48) \\ &\approx 1 - 0.9935 = 0.0065 \end{aligned}$$

Intuition

Even though daily accidents fluctuate, the total yearly sum is extremely likely to stay close to the mean (3650). An outcome of 3800 is a "2.48-sigma" event, making it quite rare ($< 1\%$).

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Problem Statement

3. In a communication system, each codeword consists of 1000 bits. Due to the noise, each bit may be received in error with probability 0.1. It is assumed bit errors occur independently. Since error correcting codes are used in this system, each codeword can be decoded reliably if there are less than or equal to 125 errors in the received codeword, otherwise the decoding fails. Using the CLT, find the probability of decoding failure.

Part (a): Modeling with Bernoulli Distribution

Why Bernoulli Distribution?

A Bernoulli trial is a random experiment with exactly two possible outcomes: "Success" (error occurs) and "Failure" (no error).

- We define $X_i = 1$ if the i -th bit is received in error.
- We define $X_i = 0$ if the i -th bit is received correctly.

From the Problem Statement:

- Each bit is independent and has an error probability $p = 0.1$.
- Individual mean: $E[X_i] = p = 0.1$.
- Individual variance: $Var(X_i) = p(1 - p) = 0.09$.

The Total Error Count

The total number of errors in a codeword of $n = 1000$ bits is the sum:

$$Y = X_1 + X_2 + \cdots + X_{1000}$$

This sum follows a Binomial distribution $B(n, p)$.

Part (b): Applying the CLT

The CLT Mechanism

The Central Limit Theorem allows us to treat the sum of many independent trials (Y) as a Normal distribution, even though the individual X_i are discrete (0 or 1).

Step-by-Step Application:

- 1 **Calculate Total Parameters:** $E[Y] = np = 100$ and $Var(Y) = np(1 - p) = 90$.
- 2 **Standardization:** We create a standard normal variable Z by subtracting the mean and dividing by the standard deviation:

$$Z = \frac{Y - 100}{\sqrt{90}} \approx N(0, 1)$$

- 3 **Transform the Threshold:** To find $P(Y > 125)$, we apply the same transformation to 125.

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Problem Statement

4. 50 students live in a dormitory. The parking lot has the capacity for 30 cars. Each student has a car with probability $\frac{1}{2}$, independently from other students. Use the CLT (with continuity correction) to find the probability that there won't be enough parking spaces for all the cars?

Part (a): Modeling Parking Demand

Why Bernoulli Distribution?

Each of the 50 students independently decides to have a car with probability $1/2$.

- Let $X_i = 1$ if student i has a car, $X_i = 0$ otherwise.
- $X_i \sim \text{Bernoulli}(1/2)$.

System Parameters ($n = 50$):

- **Individual Mean:** $EX_i = 1/2$.
- **Individual Variance:** $\text{Var}(X_i) = p(1 - p) = 1/4$.
- **Total Demand (Y):** $Y = \sum_{i=1}^{50} X_i$.
- **Total Mean:** $EY = 50 \times (1/2) = 25$.
- **Total Variance:** $\text{Var}(Y) = 50 \times (1/4) = 12.5$.

Part (b): Continuity Correction

What is Continuity Correction?

We are using a continuous distribution (Normal) to approximate a discrete one (Binomial). To bridge the gap, we shift the boundary by 0.5.

The Logic:

- The lot fails if demand $Y > 30$.
- Since Y is an integer, $Y > 30$ is the same as $Y \geq 31$.
- To be more accurate in the continuous world, we take the midpoint:

$$P(Y > 30) = P(Y \geq 31) \approx P(Y \geq 30.5)$$

Part (c): Probability Calculation

Normal Approximation

By CLT,

$$Z = \frac{Y - E[Y]}{\sqrt{\text{Var}(Y)}} = \frac{Y - 25}{\sqrt{12.5}}$$

The Calculation:

$$\begin{aligned} P(Y \geq 30.5) &= P\left(\frac{Y - 25}{\sqrt{12.5}} \geq \frac{30.5 - 25}{\sqrt{12.5}}\right) \\ &= P\left(Z \geq \frac{5.5}{\sqrt{12.5}}\right) \approx 1 - \Phi(1.56) \\ &\approx 1 - 0.94 = 0.06 \end{aligned}$$

Conclusion

There is approximately a **6% chance** that the parking lot will not have enough spaces.

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Problem Statement

5. The amount of time needed for a certain machine to process a job is a random variable with mean $EX_i = 10$ minutes and $\text{Var}(X_i) = 2$ minutes². The time needed for different jobs are independent from each other. Find the probability that the machine processes less than or equal to 40 jobs in 7 hours.

Part (a): Modeling Total Processing Time

Why Normal Approximation?

We are summing $n = 40$ independent jobs. According to the Central Limit Theorem (CLT), the total time Y will follow a Normal distribution regardless of the original distribution of X_i .

Step 1: Identify Individual Parameters For each job X_i :

$$E[X_i] = 10, \quad \text{Var}(X_i) = 2$$

Step 2: Aggregate for $n = 40$ Jobs Let $Y = \sum_{i=1}^{40} X_i$. By linearity of expectation and variance:

- **Total Mean:** $E[Y] = n \cdot E[X_i] = 40 \times 10 = 400$
- **Total Variance:** $\text{Var}(Y) = n \cdot \text{Var}(X_i) = 40 \times 2 = 80$

Part (b): Probability and Standardization

Defining the Event

"Processing ≤ 40 jobs in 7 hours" means the time for 40 jobs Y **exceeded** the 420-minute limit.

Target Probability: $P(Y > 420)$

Step 3: Standardization (Z-score) Transform the threshold 420 into the standard normal scale:

$$Z = \frac{Y - E[Y]}{\sqrt{\text{Var}(Y)}} = \frac{420 - 400}{\sqrt{80}} \approx 2.24$$

Step 4: Final Calculation

$$P(Z > 2.24) = 1 - \Phi(2.24) \approx 1 - 0.9875 = 0.0125$$

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Problem Statement

6. You have a fair coin. You toss the coin n times. Let X be the portion of times that you observe heads. How large n has to be so that you are 95% sure that $0.45 \leq X \leq 0.55$? In other words, how large n has to be so that

$$P(0.45 \leq X \leq 0.55) \geq .95 \text{ ?}$$

Part (a): Modeling the Proportion of Heads

The Random Variable

We toss a fair coin n times. Let X be the **portion** (average) of heads observed:

$$X = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{Y}{n}$$

where each $X_i \sim \text{Bernoulli}(1/2)$.

Parameters for X :

- **Mean:** $E[X] = p = 0.5$
- **Variance:** $\text{Var}(X) = \frac{p(1-p)}{n} = \frac{1}{4n}$
- **Standard Deviation:** $\sigma_X = \sqrt{\frac{1}{4n}} = \frac{1}{2\sqrt{n}}$

Part (b): Solving for n using CLT

Setting the Confidence Level

We want to be 95% sure that X is within $[0.45, 0.55]$.

$$P(0.45 \leq X \leq 0.55) \geq 0.95$$

Step 1: Standardize the interval Subtract the mean (0.5) and divide by the standard deviation ($\frac{1}{2\sqrt{n}}$):

$$P\left(\frac{0.45 - 0.5}{1/(2\sqrt{n})} \leq Z \leq \frac{0.55 - 0.5}{1/(2\sqrt{n})}\right) \geq 0.95$$

$$P(-0.1\sqrt{n} \leq Z \leq 0.1\sqrt{n}) \geq 0.95$$

Step 2: Find n using the Z-table For a 95% confidence interval, the critical value is $Z = 1.96$:

$$0.1\sqrt{n} \geq 1.96 \implies \sqrt{n} \geq 19.6 \implies n \geq 384.16$$

Conclusion: The engineer needs at least **385** tosses.

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Problem Statement

7. An engineer is measuring a quantity q . It is assumed that there is a random error in each measurement, so the engineer will take n measurements and reports the average of the measurements as the estimated value of q . Specifically, if Y_i is the value that is obtained in the i th measurement, we assume that

$$Y_i = q + X_i,$$

where X_i is the error in the i 'th measurement. We assume that X_i 's are i.i.d with $EX_i = 0$ and $\text{Var}(X_i) = 4$ units. The engineer reports the average of measurements

$$M_n = \frac{Y_1 + Y_2 + \dots + Y_n}{n}.$$

How many measurements does the engineer need to make until he is 95% sure that the final error is less than 0.1 units? In other words, what should the value of n be such that

$$P(q - 0.1 \leq M_n \leq q + 0.1) \geq 0.95 \text{ ?}$$

Transforming to Standard Normal Z

The Goal

We need the final error to be within 0.1 units of the true value q :

$$P(q - 0.1 \leq M_n \leq q + 0.1) \geq 0.95$$

Step 1: Center the Variable Subtract the mean (q) from all parts of the inequality to isolate the error:

$$P(-0.1 \leq M_n - q \leq 0.1) \geq 0.95$$

Step 2: Scale by Standard Deviation Divide all parts by the standard deviation of the average, $\sigma_{M_n} = \frac{2}{\sqrt{n}}$:

$$P\left(\frac{-0.1}{2/\sqrt{n}} \leq \frac{M_n - q}{2/\sqrt{n}} \leq \frac{0.1}{2/\sqrt{n}}\right) \geq 0.95$$

Step 3: Substitute Z Since $\frac{M_n - q}{\sigma_{M_n}} \approx Z \sim N(0, 1)$, we get:

$$P(-0.05\sqrt{n} \leq Z \leq 0.05\sqrt{n}) \geq 0.95$$

Solving for n using CLT

Step 3: Apply CLT for Standardization

By the Central Limit Theorem, for large n , the standardized average is approximately standard normal:

$$Z = \frac{M_n - q}{2/\sqrt{n}} \sim N(0, 1)$$

Transformation of the Inequality: We convert the error bound of 0.1 into Z -score units:

$$P(-0.05\sqrt{n} \leq Z \leq 0.05\sqrt{n}) \approx 2\Phi(0.05\sqrt{n}) - 1$$

Step 4: Solve for the Minimum n Setting the probability to at least 0.95:

- $2\Phi(0.05\sqrt{n}) - 1 \geq 0.95 \implies \Phi(0.05\sqrt{n}) \geq 0.975$
- From the Z -table: $0.05\sqrt{n} \geq 1.96$
- $\sqrt{n} \geq 39.2 \implies n \geq 1536.64$

Conclusion: The engineer needs at least **1537** measurements.